

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Data Structure

Time: 1.5 Hours

Semester / Programme: sem 3

Full Marks: 20

Instructions: Answer Q1 and any 3 from Q2-Q5

SECTION - A (Compulsory)

1(a) What is the time complexity of binary search?

[1]

$$\mathcal{O}(\log n)$$

SECTION - B

Answer any 3

Q.2. (a) Explain array storage representation.

[2.5]

$$A[i] = B + i \times w$$

(b) Derive stack overflow condition.

[2.5]

--- END OF PAPER ---